Mahesh Kumar

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EDUCATION

Spoorthy Degree and PG College, *Bachelor of Science* 2022 – 2025
Data Science : 8.6 GPA

Narayana Junior College, *Intermediate Education* 2020 – 2022
MPC (Maths, Physics, Chemistry) : 8.1 GPA

SKILLS

MLOps & Automation: MLflow, Optuna, Git, GitHub Actions, SHAP, Gradio, Streamlit

ML Frameworks & Libraries: Python, PyTorch, TensorFlow, Keras, Scikit-Learn, CatBoost, XGBoost, LightGBM, RAPIDS (cuDF/cuML), Hugging Face, YOLOv8, OpenCV, SQL, Pandas, NumPy

Development & Environment: Bash, Jupyter Notebook, Google Colab, VS Code

Core Competencies: End-to-End ML Systems, Model Productionization, Data-Driven Optimization, GPU Acceleration

PROJECTS

Aerial Disaster Response AI: Real-Time Computer Vision

- Built and deployed an end-to-end Computer Vision web application on Streamlit Community Cloud to detect building damage in high-resolution satellite imagery for disaster response.
- Overcame small-object detection limitations by integrating YOLOv8 with SAHI (Slicing Aided Hyper Inference), boosting small-object detection accuracy (mAP) by over 18% compared to baseline models.
- Streamlined the MLOps deployment pipeline by automating model weight retrieval, dynamic dependency generation, and class merging to stabilize production under Google Colab's 15GB GPU memory constraints.

Swiggy ETA Prediction: GPU-Accelerated Tabular ML

- Engineered a high-performance ETA prediction pipeline using LightGBM and XGBoost, processing over 700,000 rows of geospatial data with GPU-accelerated RAPIDS (cuDF/cuML) to eliminate CPU bottlenecks.
- Implemented a rigorous Out-Of-Fold (OOF) Target Encoding strategy and K-Fold cross-validation to prevent data leakage, achieving a highly robust R^2 score of 0.84 on unseen test data.
- Designed domain-specific features (Haversine distance, bearing) and automated the hyperparameter optimization pipeline over 50 trials using Optuna to minimize MSE to 4.12 minutes.

AI-Firewall: Deep Learning Network Intrusion Detection System

- Architected a Deep Learning classification system using TensorFlow/Keras to analyze complex network traffic patterns, processing 500,000+ network packets to detect Zero-Day cyber attacks with 98.5% overall accuracy.
- Resolved severe class imbalance through strategic data preprocessing (StandardScaler, OneHotEncoder), ensuring stable gradient descent and achieving a 94% recall on minority attack classes.
- Optimized GPU memory management and training stability by implementing dynamic batching, Dropout regularization, and EarlyStopping callbacks to prevent model overfitting.

ACHIEVEMENTS & COMPETITIONS

Amazon ML Challenge , *Top 1.2% Competitor*

- Secured a Top 1.2% final rank (967 of 82,000+ participants) by architecting and deploying a state-of-the-art regression system for a highly competitive retail price prediction task.
- Reduced the baseline error metric by over 35% (from 79.0 to 50.6) through systematic feature engineering, target variable log-transformation, and comprehensive model benchmarking.
- Constructed the champion model from a weighted ensemble of a PyTorch MLP and a CatBoost regressor, which surpassed all standalone models by effectively correcting for individual prediction errors.